MLPerf Storage v2.0 System Architecture

Dell PowerEdge R7525 Gen4 system, local storage submission

Submitter: Kingston Digital, Inc. (Hazem Awadallah) | Date: July 2025 | Division: CLOSED

System Specifications

Component	Specification
System Model	Dell PowerEdge R7525 Gen4 system, local storage submission
CPU	AMD EPYC 7302 16-Core Processor (32 cores)
Memory	251 GB
Operating System	Ubuntu 22.04.5 LTS
Kernel Version	6.5.0-15-generic
Storage Interface	BLOCK
Storage Location	Local

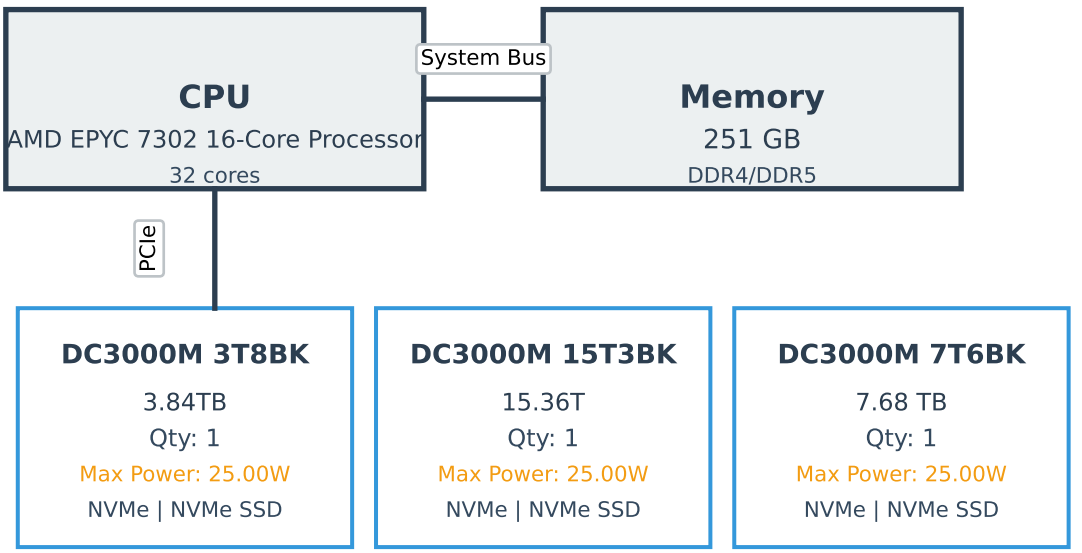
Storage Configuration

Model	Capacity	Quantity	Type	Protocol
DC3000M 3T8BK	3.84TB	1	NVMe SSD	NVMe
DC3000M 15T3BK	15.36T	1	NVMe SSD	NVMe
DC3000M 7T6BK	7.68 TB	1	NVMe SSD	NVMe

Performance Specifications

Operation	Performance
Sequential Read	14GB/s
Sequential Write	10 GB/s
Random Read	2800K IOPS
Random Write	500K IOPS

System Architecture Diagram



Interconnect: PCIe Gen5x4 connected to a Gen4x4 backplane

System Capabilities

Capability	Value
Multi-host Support	False
Simultaneous Write	False
Simultaneous Read	False
Rack Units	2U

Power Requirements & Configuration

Power Consumption Analysis

Component	Qty	PSU1 Nameplate	PSU2 Nameplate	PSU Config	Design Power	Rack Units
System (DLIO Client + Storage)	1	1200W	1200W	1A + 1P	1200W	2U
NVMe: DC3000M 3T8BK	1	25W	-	Active Only	25W	0U
NVMe: DC3000M 15T3BK	1	25W	-	Active Only	25W	0U
NVMe: DC3000M 7T6BK	1	25W	-	Active Only	25W	0U
SYSTEM TOTAL					1225W	2U

Notes:
• PSU Configuration: A = Active, P = Passive
• Design power calculation excludes passive PSUs
• NVMe power consumption based on SMART data analysis
• All power values represent maximum operational requirements
• Total rack space includes all system components

System Tuning & Configuration

MPI Configuration

Parameter	Value
MPI Version	Open MPI 4.1.2
Environment Variables	reader.read_threads=4 for resnet, reader.read_threads=16 for unet3d, cosmoflow and checkpointing

System Tuning Parameters

Parameter 1:

NVMe interrupt coalescing set to feature-id 0x8 = 0x070a (700 Âµs timeout, 11-I/O threshold) reduces CPU overhead under bursty loads.

Parameter 2:

ResNet-50 (random small reads): max_sectors_kb=32, read_ahead_kb=32, nomerges=2, 16 reader threads, optimizes random I/O throughput.

Parameter 3:

UNet3D, CosmoFlow, and checkpointing (sequential workloads): max_sectors_kb=128, read_ahead_kb=4096, nomerges=0, leverages large sequential tunables for steady bandwidth.

Parameter 4:

System-wide settings: rq_affinity=2 (NUMA-aware interrupt routing), wbt_lat_usec=0 (disable write-back throttling), add_rand (scatter I/O to avoid alignment hotspots).

Operating System Configuration

Parameter	Value
Operating System	Ubuntu 22.04.5 LTS
Release Date	2022-04-12
Kernel Version	6.5.0-15-generic
CPU Architecture	x86_64